

Algorithms

page 1

$$- x^n = x^{n/2} \cdot x^{n/2} \cdot x$$

if n odd

$$\text{return } (\text{pow}(x, n/2))^2 \cdot x$$

else

$$\text{return } (\text{pow}(x, n/2))^2$$

$$- \sum_{i=1}^n (2i+3) = 2n^2 + 3n \quad - \text{Wrong}$$

$$\sum_{i=1}^n 2i + \sum_{i=1}^n 3 = 2 \sum_{i=1}^n i + \sum_{i=1}^n 3$$

$$\frac{2(n(n+1))}{2} + 3n$$

$$- \sum_{i=0}^n 3^i = \frac{3^{n+1} - 1}{2}$$

$$\#7a \quad \sum_{i=1}^n (3i+7)$$

$$\#7f \quad \sum_{i=7}^n 4i$$

$$\#7b \quad \sum_{i=1}^n (i^2 - 2i)$$

$$\log x = \log_2 x$$

$$\log x = \log_{10} x$$

$$\#7c \quad \sum_{i=7}^n i$$

$$\ln x = \log_e x$$

$$\#7d \quad \sum_{i=5}^n (2i^2 + i)$$

$$\log_a b = \frac{\log_c b}{\log_c a}$$

$$\log_3 10 = \frac{\log_{10} 10}{\log_{10} 3}$$

$$\#7e \quad \sum_{i=1}^n 6^i$$

$$\log 255 = \frac{\log_{10} 255}{\log_{10} 2}$$

$$\log(m \cdot n) = \log m + \log n$$

$$\log \frac{m}{n} = \log m - \log n$$

$$\log n^2 = \log n \cdot \log n = 2 \log n$$

- smallest growth of a function is constant
- logs are next slowest growing
- logs all grow at the same rate
- n to a power is next, the bigger n is, the faster it grows
- exponential
- factorial

$$\begin{array}{lcl}
 0 & < & \\
 0 & \leq & \\
 0 & = & \\
 \wedge & \geq & \\
 \cap & > &
 \end{array}$$

pg. 22 order from highest to lowest

#1	2^n	$n - n^2 + 5n^3$	$(\frac{3}{2})^n$
	$\lg n$	n^3	
	$\frac{n}{2}$	$\sqrt{n}$	
	$(\lg n)^2$	n	
	$n!$	$n^3 + \lg n$	
	$\lg \lg n$	$2^n - 1$	
		$n \lg n$	
		6	

pg. 23 determine if $f \in O(g)$ or $g \in O(f)$

#2	a	$\frac{n^2 - n}{2}$	$6n$
	b	$n + 2\sqrt{n}$	n^2
	c	$n + \log n$	$n\sqrt{n}$
	d	$n^2 + 3n + 4$	n^3
	e	$n \log n$	$\frac{n\sqrt{n}}{2}$
	f	$n + \log n$	$(\sqrt{n})^2$
	g	$2(\log n)^2$	$\log n + 1$
	h	$4n \log n + n$	$\frac{n^2 - n}{2}$